

COMP6237 Data Mining

Lecture 10: Semantic Spaces (Finding Features I)

Zhiwu Huang

Zhiwu.Huang@soton.ac.uk

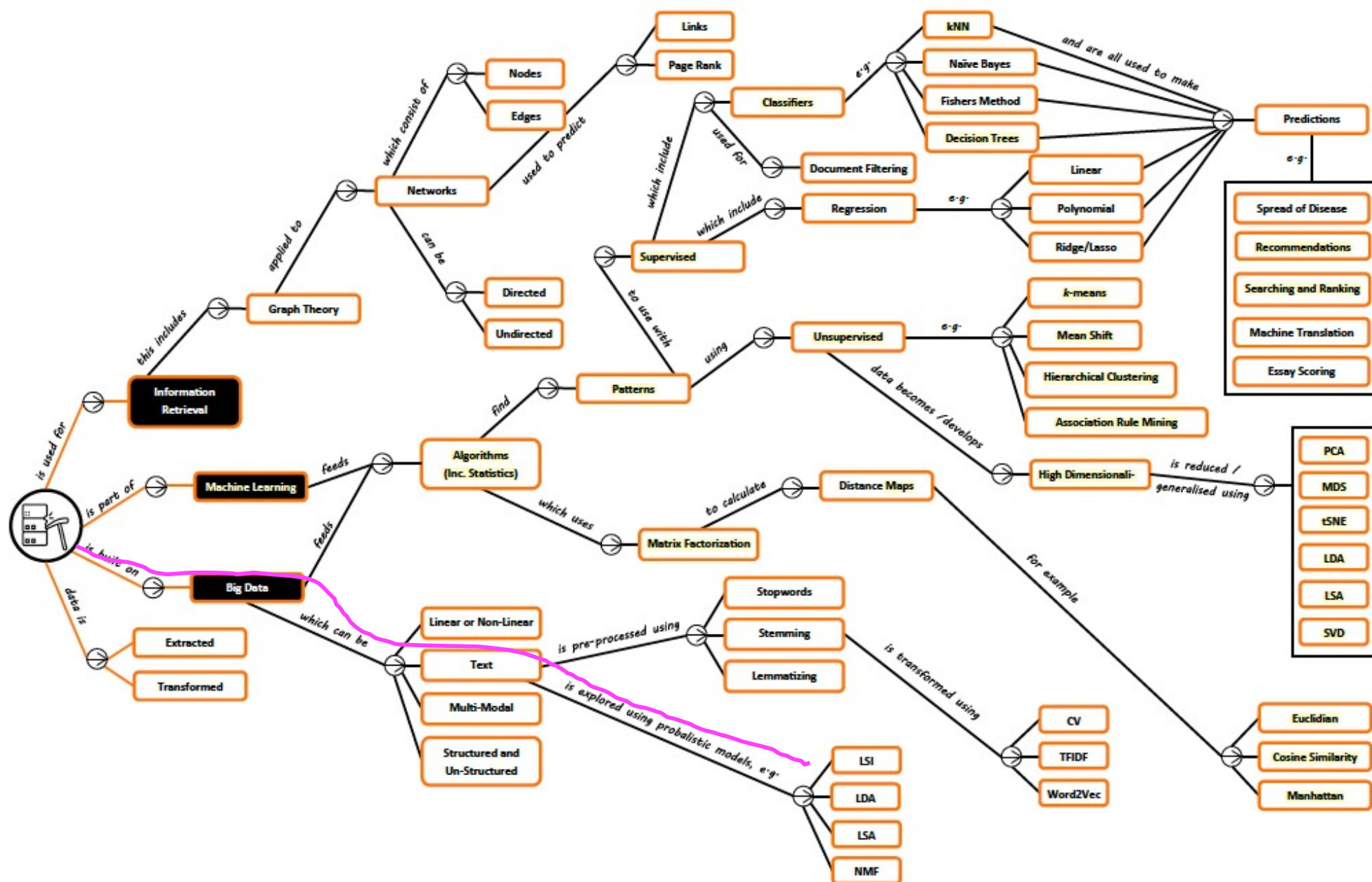
Lecturer (Assistant Professor) @ VLC of ECS
University of Southampton

Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Semantic Spaces – Roadmap



Finding Features I – Textbook

CHAPTER 10

Finding Independent Features

Most of the chapters so far have focused primarily on *supervised* classifiers, except Chapter 3, which was about *unsupervised* techniques called *clustering*. This chapter will look at ways to extract the important underlying features from sets of data that are not labeled with specific outcomes. Like clustering, these methods do not seek to make predictions as much as they try to characterize the data and tell you interesting things about it.

- ▶ Programming Collective Intelligence: Building Smart Web 2.0 Applications *T. Segaran*.

Semantic Spaces – Overview (1/4)

Distributional Semantics - Hypothesis:

Words that have similar distributions have similar meanings

"Words that occur in similar contexts have similar meanings"

Wittgenstein 1953

"A word is characterised by the company it keeps" Firth 1958

We can exploit this to uncover *hidden meanings*

Semantic Spaces – Overview (2/4)

Semantic Spaces:

- ▶ represent word meanings as vectors that keep track of the words distributional history
- ▶ focus on semantic similarity
- ▶ similarity measured using geometrical methods

e.g. Cosine similarity between **PC** and **Windows** = 0.77

Cosine similarity between **PC** and **window** = 0.13

In Japanese, A. Utsumi, IEEE SMC 2010

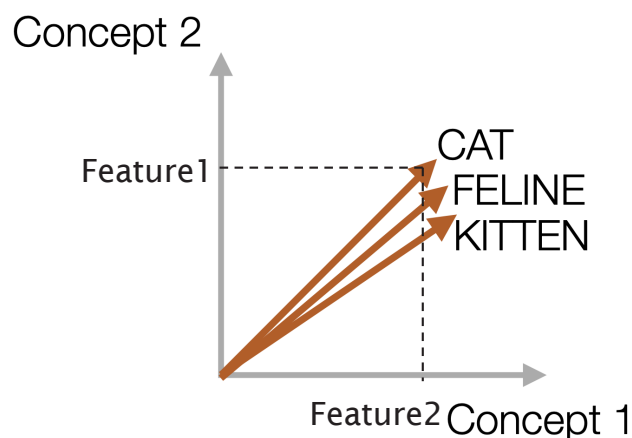


Diagram illustrating the matrix factorization process:

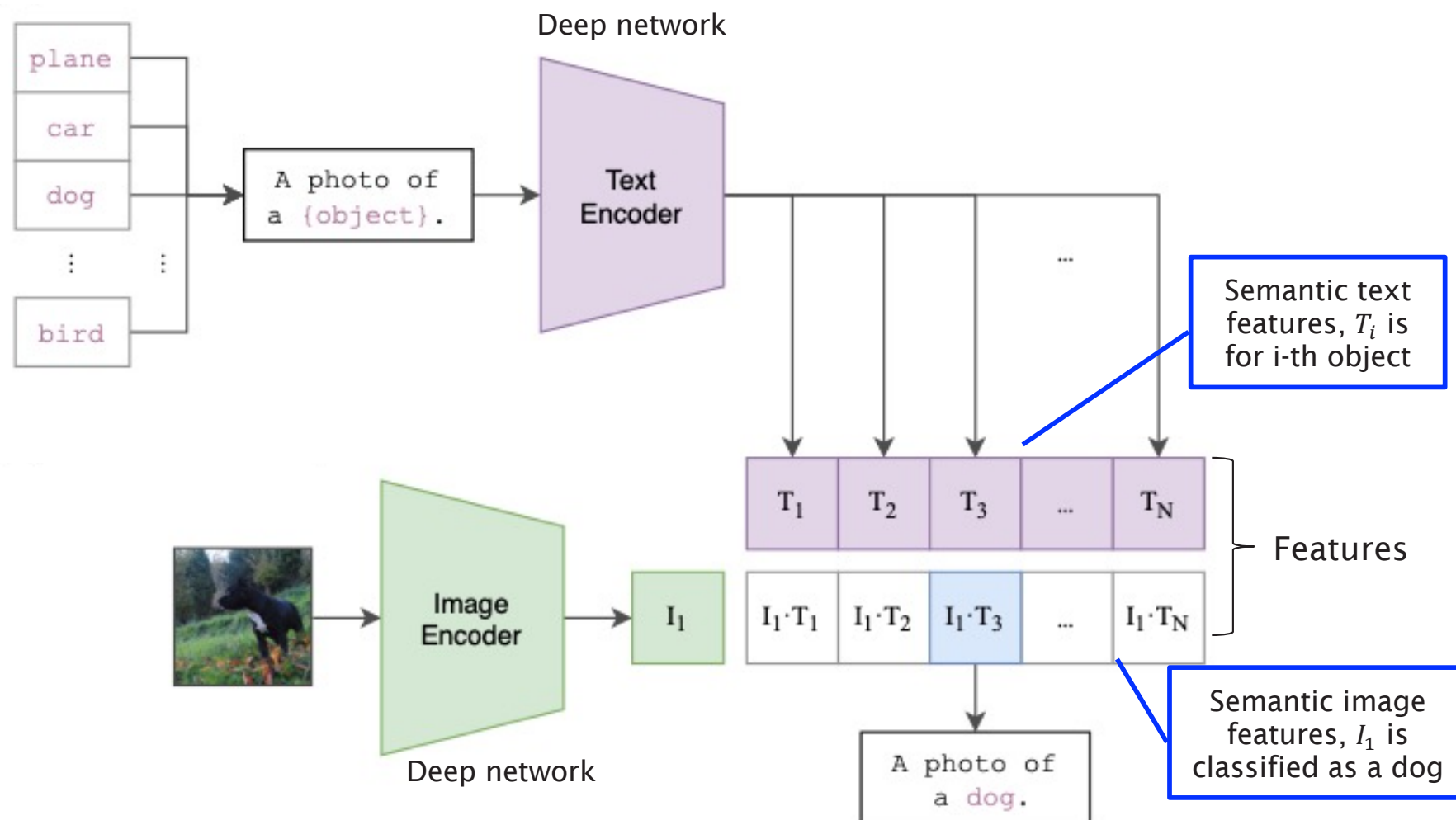
- Matrix A (dimensions $m \times n$) is labeled "docs" (vertical) and "words" (horizontal).
- Matrix U_r (dimensions $m \times r$) is labeled "words" (vertical) and "m x r" (horizontal).
- Matrix Σ_r (dimensions $r \times r$) is labeled "concepts" (vertical) and "r x r" (horizontal).
- Matrix V_r^T (dimensions $r \times n$) is labeled "docs" (vertical) and "r x n" (horizontal).

- ▶ This means it is proportional to the covariance or correlation between the documents
- ▶ These are the *concepts*

6/34

Semantic Spaces – Overview (4/4)

Contrastive Language-Image Pre-training (CLIP) uses an abundantly available source of supervision: the text paired with images found across the internet



Semantic Spaces – Learning Outcomes

- **LO1:** Demonstrate an understanding of techniques for finding independent semantic features, such as: (exam)
 - ❖ Comprehending the core concepts of Latent Semantic Analysis (LSA) and apply LSA on a dataset
 - ❖ Understanding the key pipeline of Contrastive Language-Image Pre-Training (CLIP)
 - ❖ Discussing the advantages and disadvantages of algorithms like LSA and CLIP
- **LO2:** Implement the learned algorithms for independent semantic feature learning (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

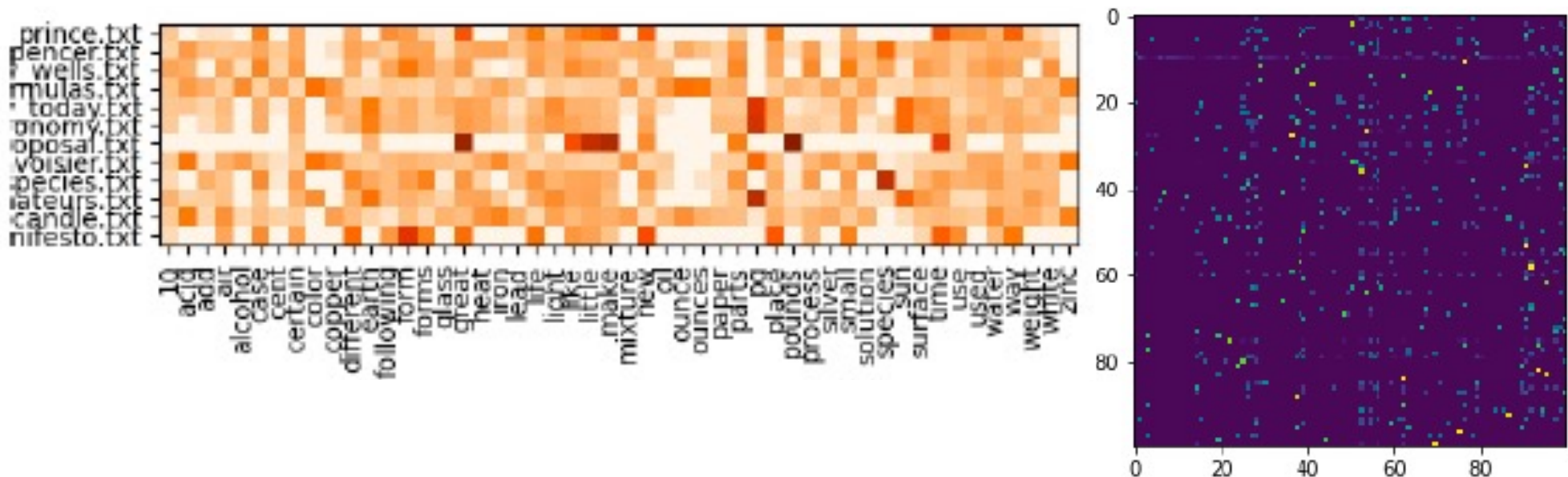
- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*

Semantic Spaces – Latent Semantic Analysis

Matrix Construction:

Consider a term-document matrix which describes occurrences of terms in documents

- ▶ Sparse
- ▶ Weighted (e.g. TF.IDF)

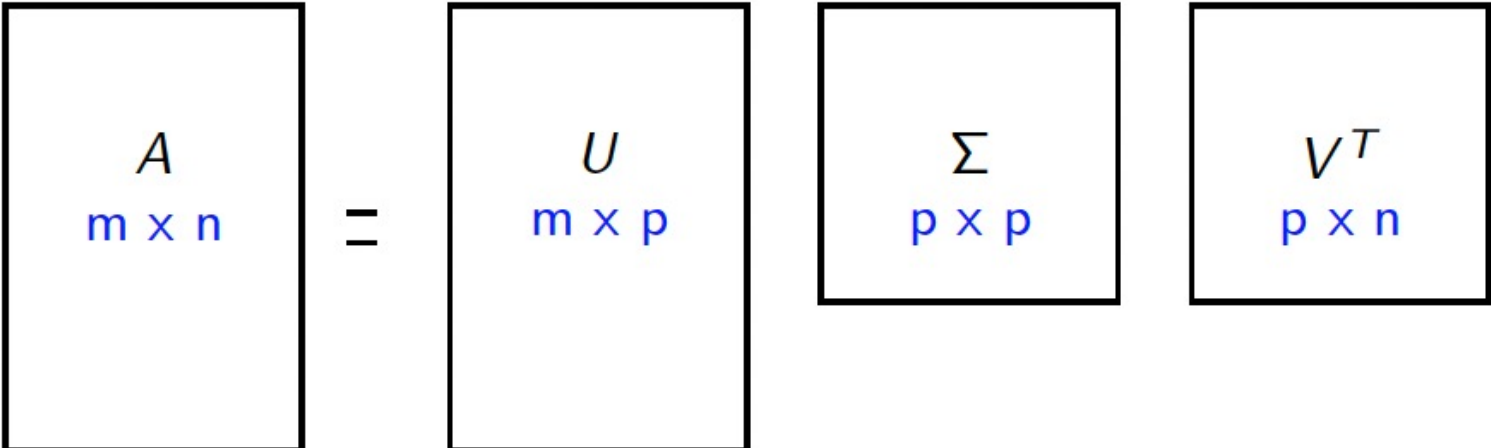


Semantic Spaces – Latent Semantic Analysis

Latent Semantic Analysis (LSA) makes a low-rank approximation
It assumes the term-document matrix:

- ▶ is noisy, and should be de-noised
- ▶ is more sparse than it should be

Semantic Spaces – Recap SVD

$$A = U \Sigma V^T$$


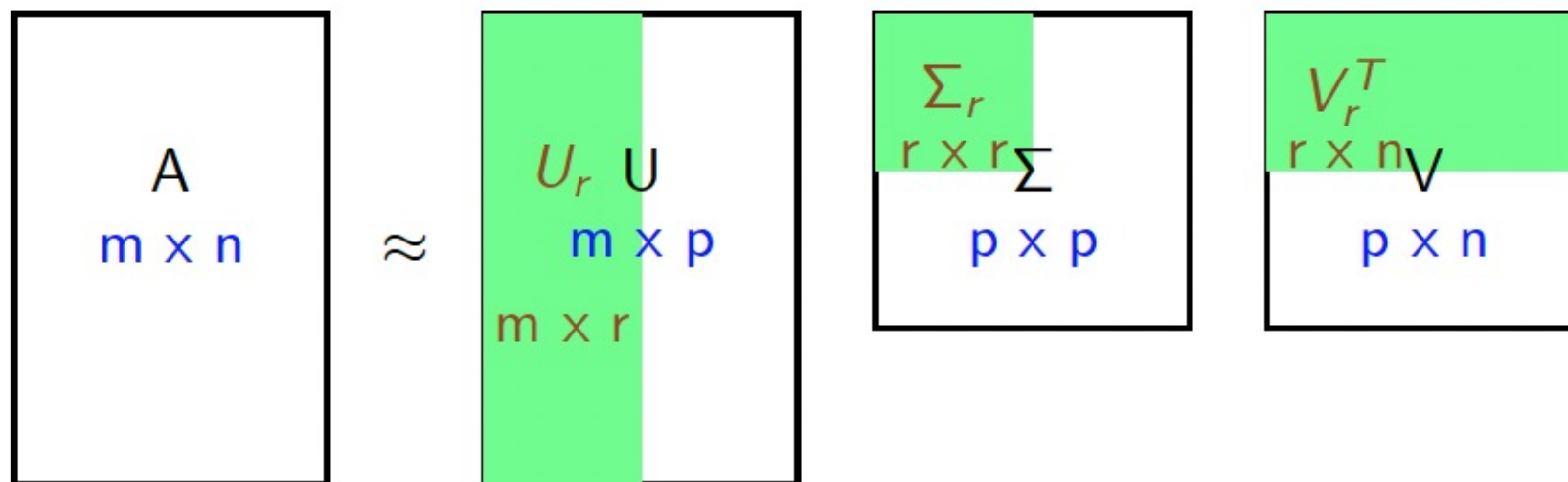
The diagram shows the SVD decomposition $A = U \Sigma V^T$. Matrix A is $m \times n$. Matrix U is $m \times p$. Matrix Σ is $p \times p$. Matrix V^T is $p \times n$. The matrices are represented by boxes with their dimensions written below them.

Where p is rank of matrix A

U called *left singular vectors*, contains the eigenvectors of AA^T ,
 V called *right singular vectors*, contains the eigenvectors of $A^T A$
 Σ contains square roots of eigenvalues of AA^T and $A^T A$

If A is matrix of mean centred featurevectors, V contains principal components of the covariance matrix

Semantic Spaces – Recap Truncated SVD

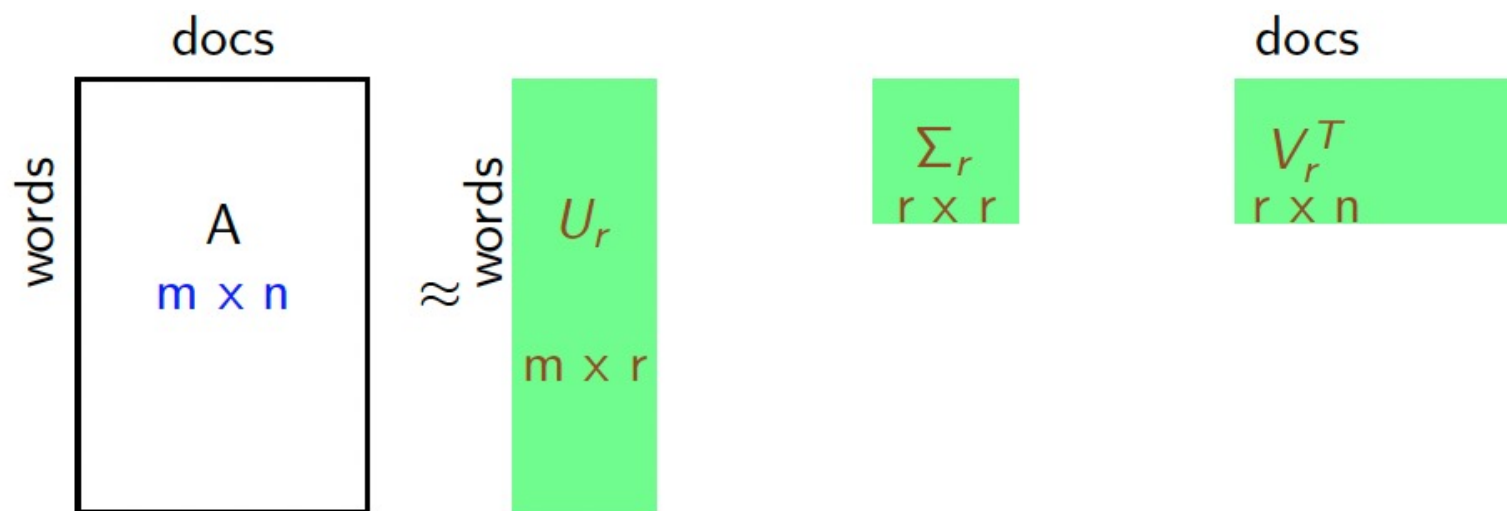


Uses only the largest r singular values (and corresponding left and right vectors)

This can give a *low rank approximation* of A , $\tilde{A} = U_r \Sigma_r V_r$

This has the effect of minimising the Frobenius norm of the difference between A and $\tilde{A}$

Semantic Spaces – Recap Truncated SVD



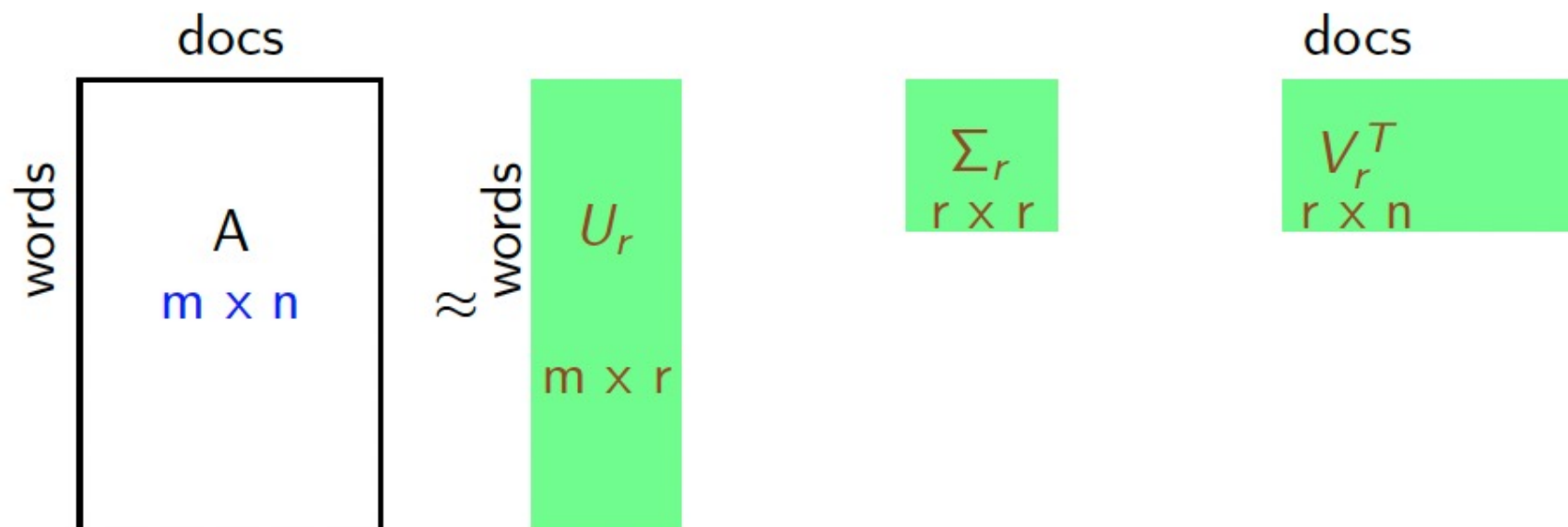
Each row of V_r corresponds to an eigenvector of $A^T A$

- ▶ This means it is proportional to the covariance or correlation between the documents
- ▶ These are the *concepts*

Each row of U_r describes a term as a vector of weights with respect to r *concepts*

Each column of V_r describes a document as a vector of weights with respect to r *concepts*

Semantic Spaces – LSA



Term concepts and document concepts have the same dimensionality, but represent different spaces.

Semantic Spaces – LSA

Example:

a set of strings:

m1 "Human machine interface for ABC computer applications"

m2 "A survey of user opinion of computer system response time"

m3 "The EPS user interface management system"

m4 "System and human system engineering testing of EPS"

m5 "Relation of user perceived response time to error measurement"

g1 "The generation of random, binary, ordered trees"

g2 "The intersection graph of paths in trees"

g3 "Graph minors IV: Widths of trees and well-quasi-ordering"

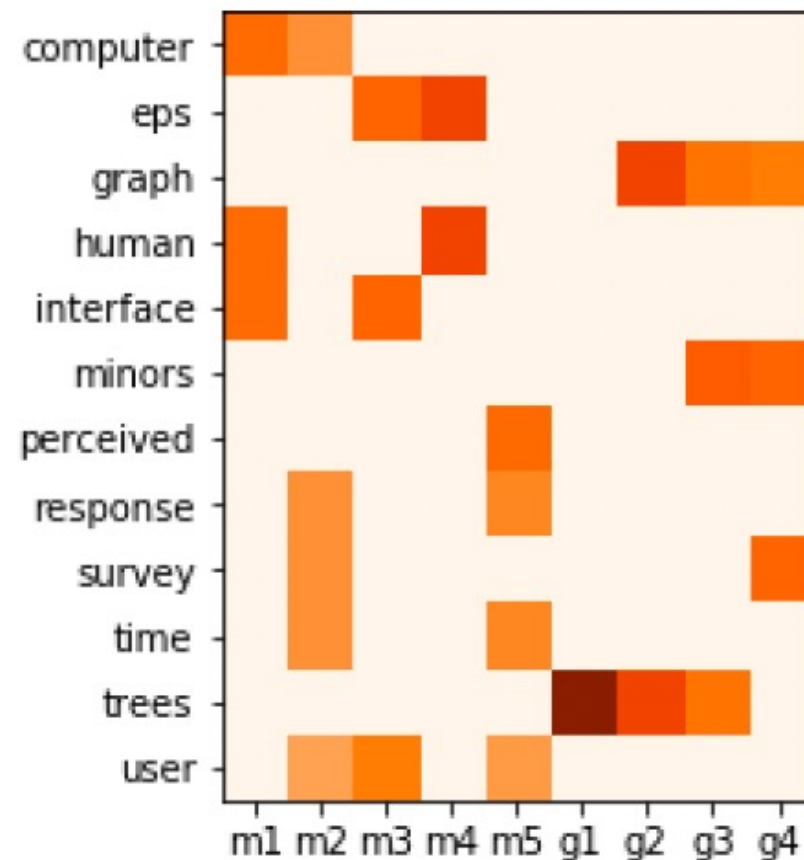
g4 "Graph minors: A survey"

<http://lsa.colorado.edu/papers/dp1.LSAintro.pdf>

Semantic Spaces – LSA

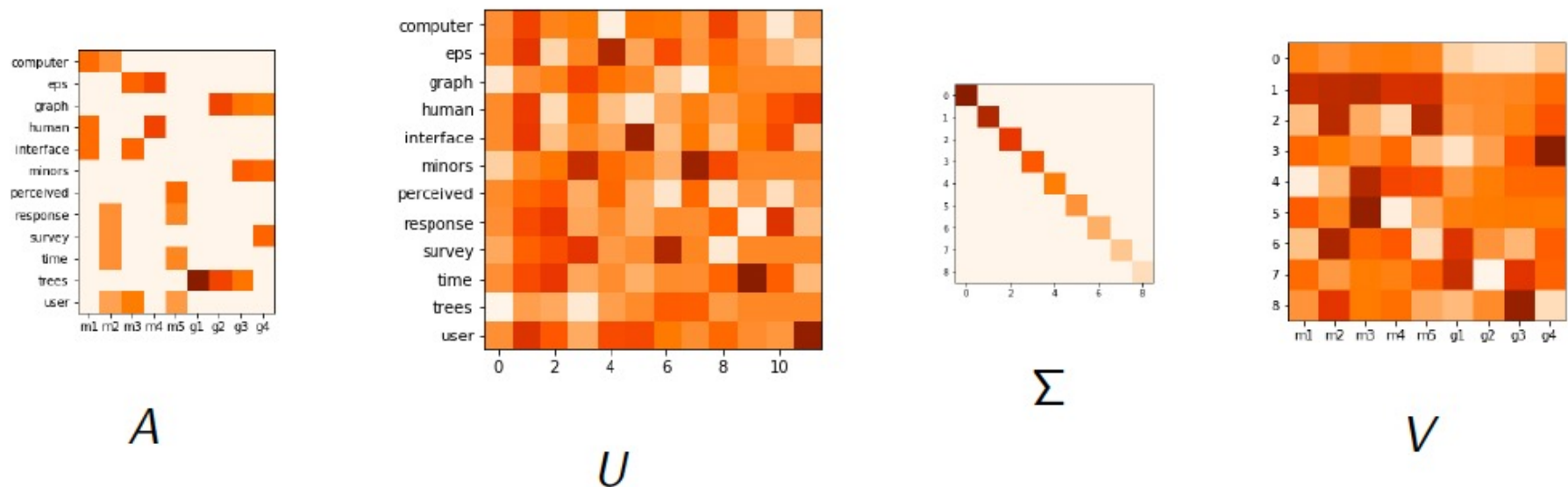
calculate TF.IDF

0.58	0.46	0.	0.	0.	0.	0.	0.	0.
0.	0.	0.6	0.71	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.71	0.55	0.52
0.58	0.	0.	0.71	0.	0.	0.	0.	0.
0.58	0.	0.6	0.	0.	0.	0.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.63	0.6
0.	0.	0.	0.	0.58	0.	0.	0.	0.
0.	0.46	0.	0.	0.49	0.	0.	0.	0.
0.	0.46	0.	0.	0.	0.	0.	0.	0.6
0.	0.46	0.	0.	0.49	0.	0.	0.	0.
0.	0.	0.	0.	0.	1.	0.71	0.55	0.
0.	0.4	0.52	0.	0.43	0.	0.	0.	0.



Semantic Spaces – LSA

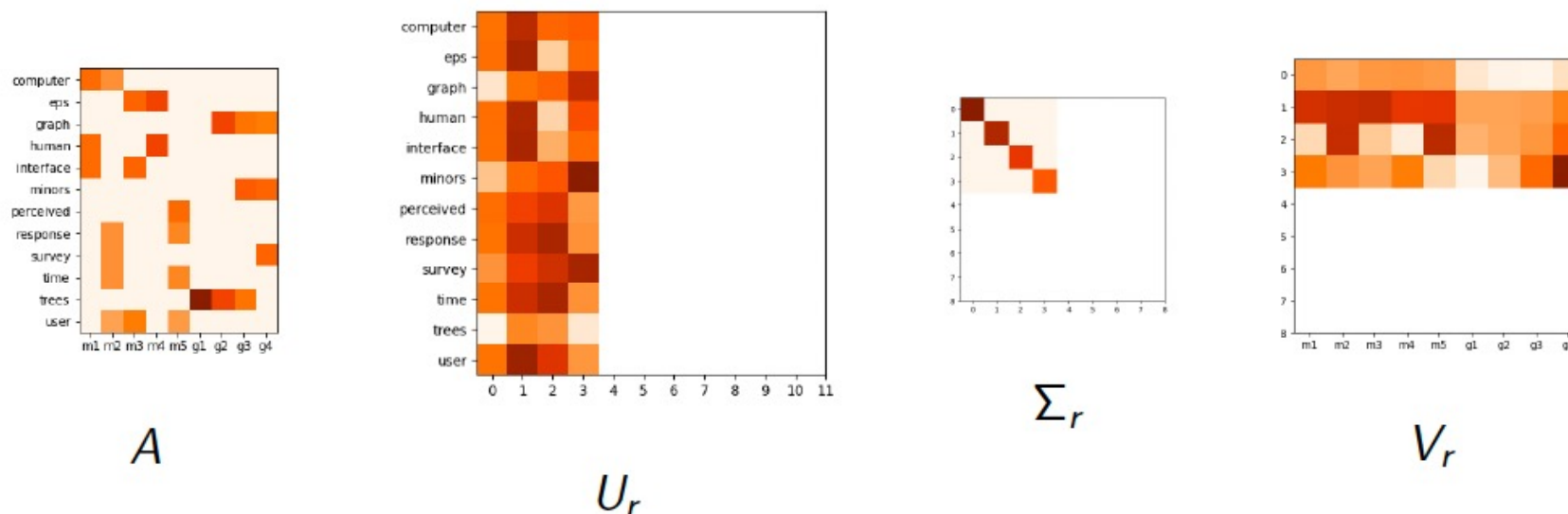
$$\text{SVD: } A = U\Sigma V^T$$



We then reduce the dimensionality by choosing only the first few eigenvalues (in Σ) and the corresponding columns in U and V .

Semantic Spaces – LSA

$$\text{SVD: } A \approx U_r \Sigma_r V_r^T \quad r = 4$$



Each row of U_r describes a word as a vector of weights with respect to r concepts

Each column of V_r describes the title as a vector of weights with respect to r concepts

Semantic Spaces – LSA

What do these 'concepts' mean?

U_r gives us the weighting of the words for each concept

We can show that weighting for each concept:

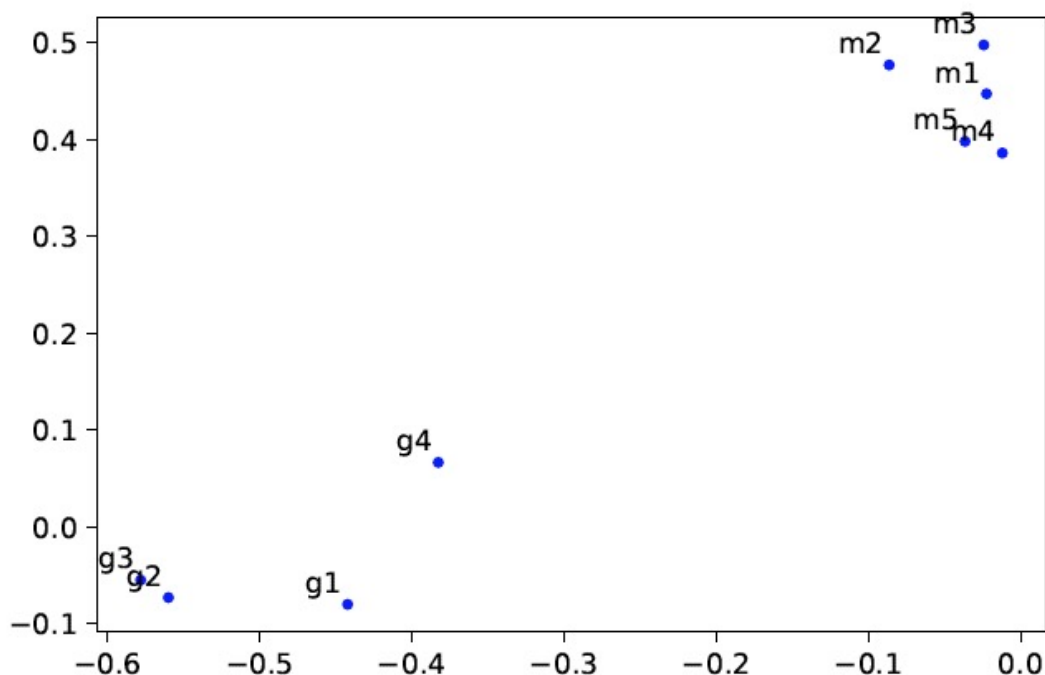
different	different	different	different
copper	copper	copper	copper
color	color	color	color
certain	certain	certain	certain
cent	cent	cent	cent
case	case	case	case
alcohol	alcohol	alcohol	alcohol
air	air	air	air
add	add	add	add
acid	acid	acid	acid
10	10	10	10
Topic: 0	1	2	3

This shows us perhaps that the concepts are not always particularly meaningful.

Semantic Spaces – LSA

Cosine similarity of document vectors can be compared ($r = 2$)

- ▶ Vectors for “m1” and “m2” give cosine similarity = 0.93
- ▶ Vectors for “g1” and “g2” give cosine similarity = 0.83
- ▶ Vectors for “g1” and “m1” give cosine similarity = 0.18

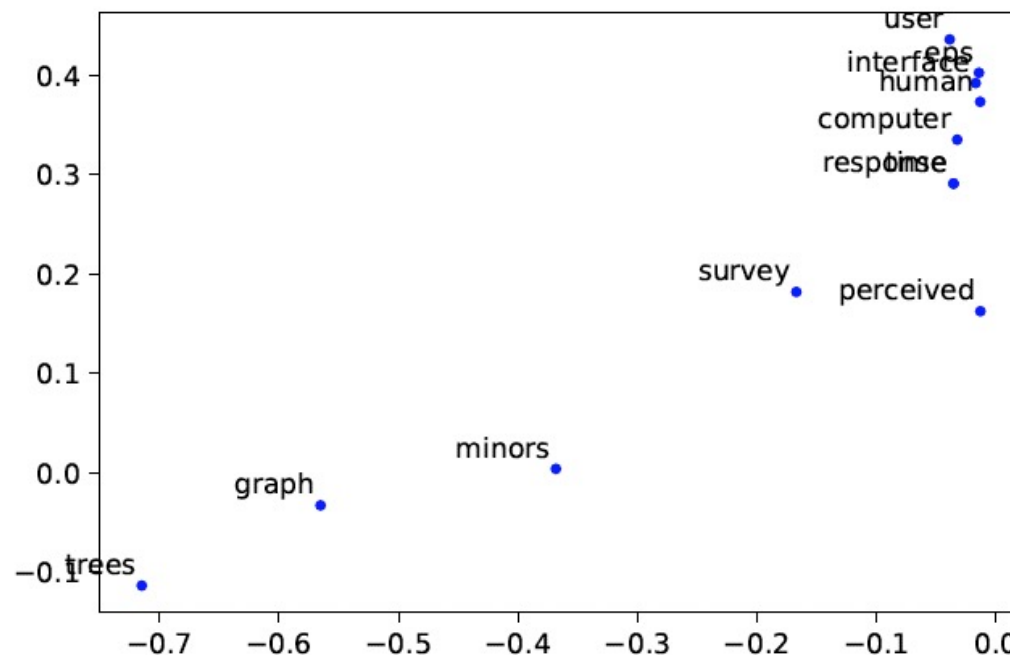


Clustering algorithms can be used on the vectors

Semantic Spaces – LSA

Cosine similarity of word vectors can be compared ($r = 2$)

- ▶ Vectors for “human” and “interface” give cos similarity = 0.95
- ▶ Vectors for “human” and “user” give cos similarity = 0.11
- ▶ Vectors for “graph” and “minor” give cos similarity = 0.90



Clustering algorithms can be used on the vectors

Semantic Spaces – LSI

Latent Semantic Indexing (LSI) LSA can be used for document retrieval

- ▶ Given Query: view as query vector $\mathbf{q}$
- ▶ Project $\mathbf{q}$ in to document space
- ▶ Compare with document vectors, find closest

Results work mathematically

However, results may not be easy to interpret in terms of natural language.

Semantic Spaces – LSA

Problems?

- ▶ Polysemious words - with multiple meanings - aren't captured
 - ▶ The vector representation averages all meanings of the word
 - ▶ e.g. 'fit' is an adjective and a verb
- ▶ Word order is ignored (use n-grams?) An n-gram is a sequence of n words
- ▶ LSA assumes words and documents form a joint Gaussian distribution, however a Poisson distribution is observed

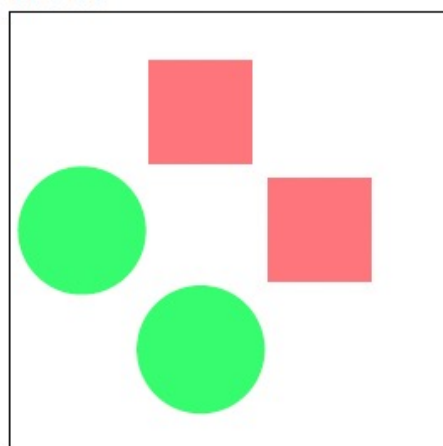
Semantic Spaces – LSA

So far: *bag of words* (BOW) from natural language.

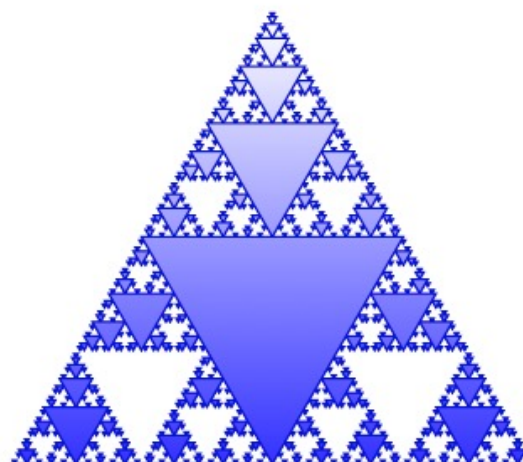
However the maths should work for compositions of occurrences in any unit.

For example, in image search we might want to search for other images with circles.

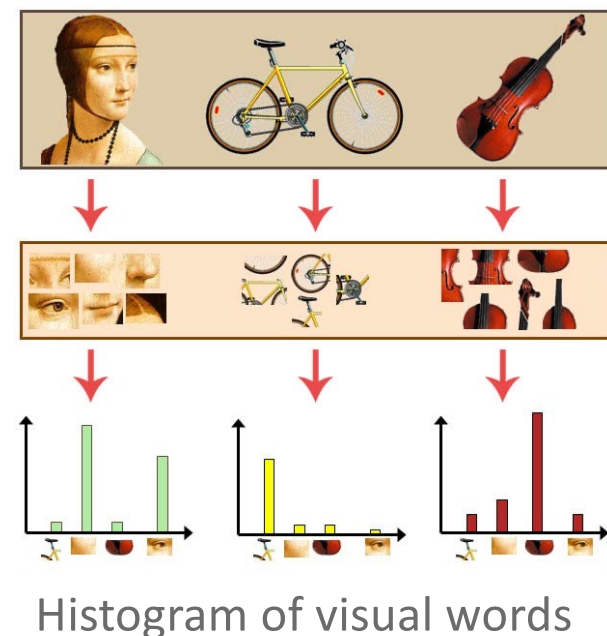
The image could be encoded with the number of different shapes it has.



○	△	□	×	*
2	0	3	0	0



○	△	□	×	*
0	∞	0	0	0



Semantic Spaces – LSA

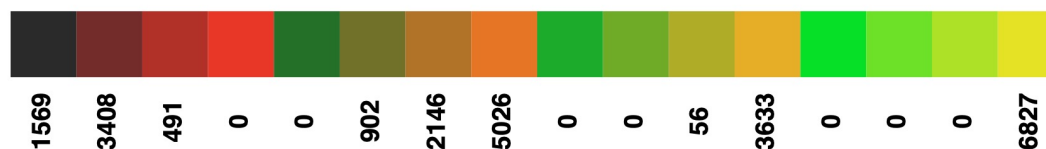
Need to make a large multidimensional space in which images, keywords and visual terms can be placed

In training:

- ▶ Learn how images and keywords are related
- ▶ Place images and keywords close together in the space

Unannotated images can be placed in the space based on the visual terms they contain

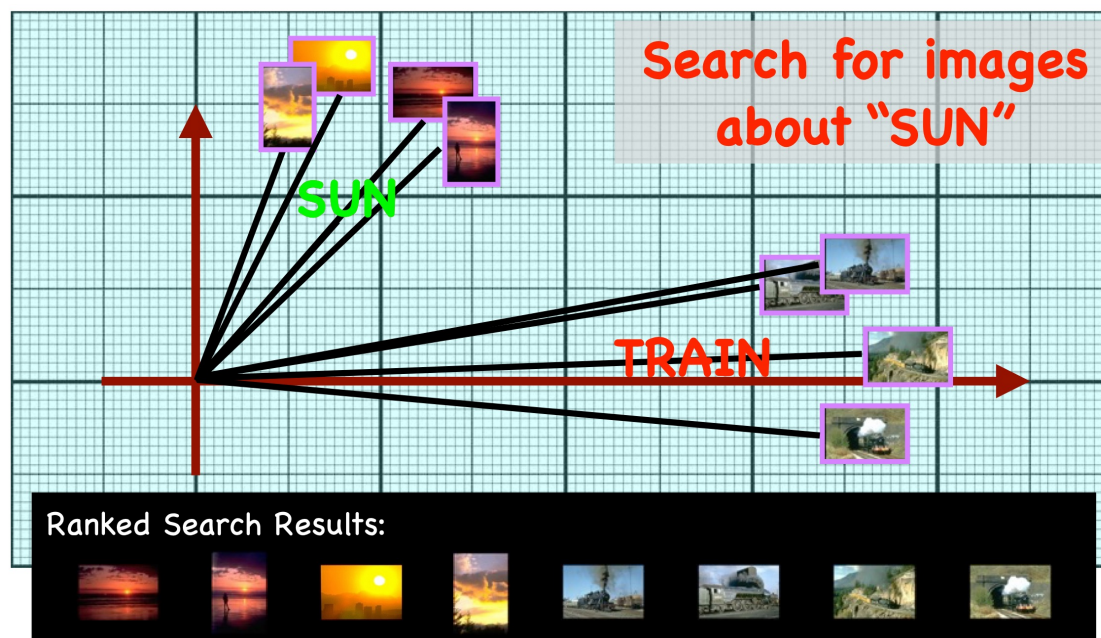
- ▶ Images can be placed based on their visual terms in the space
- ▶ They should lie near the keywords that describe them



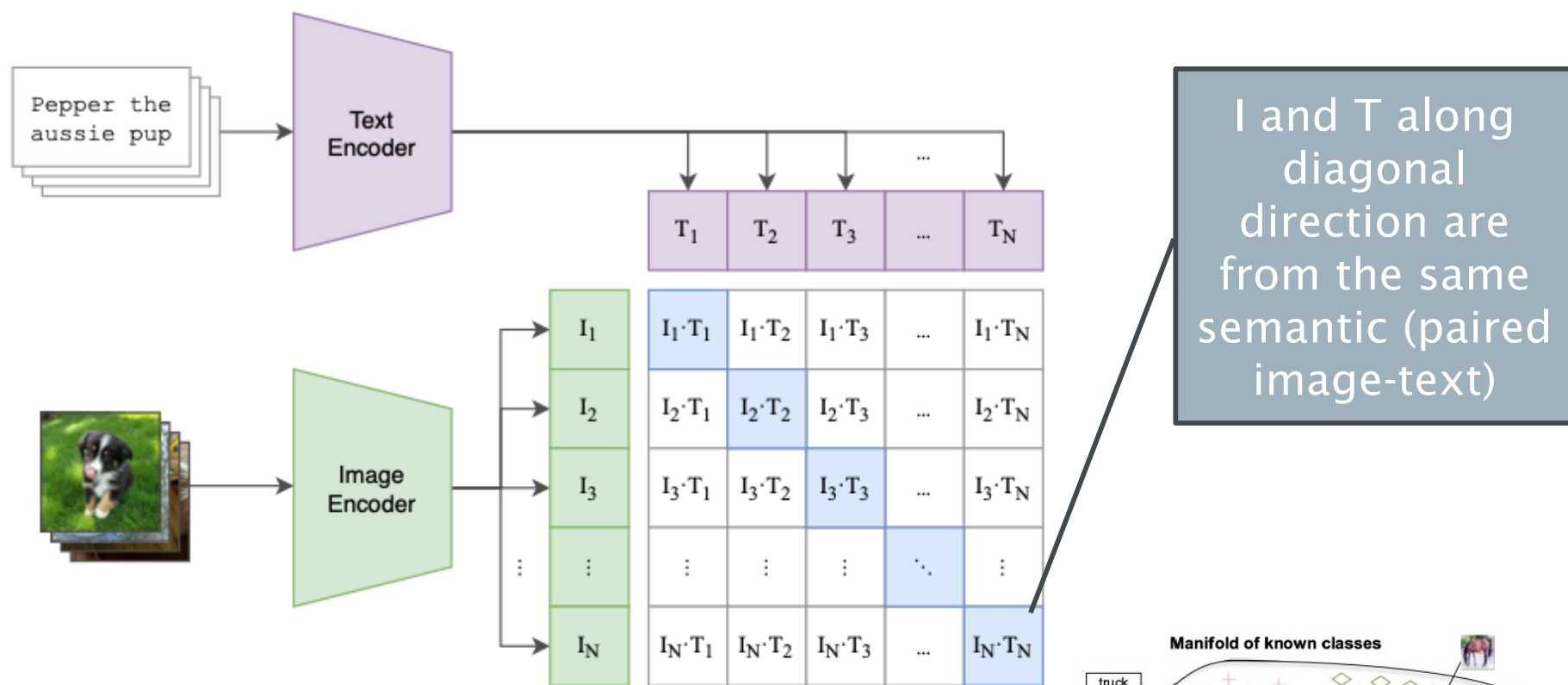
Semantic Spaces – LSA

This lower dimensional space can be used to:

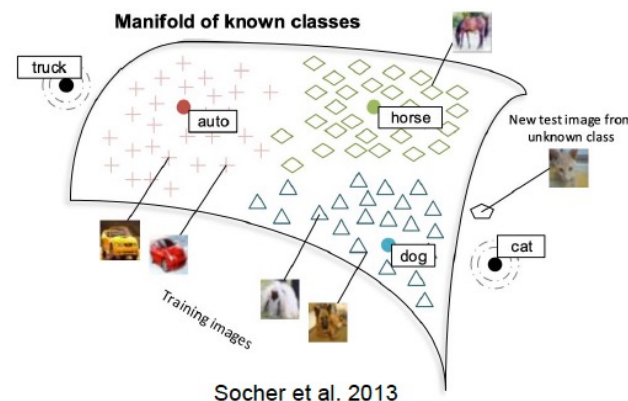
- ▶ Find Images using similar words
- ▶ Find images with similar images
- ▶ Return possible key words for an image
- ▶ Find relationships between words, and between words and visual terms



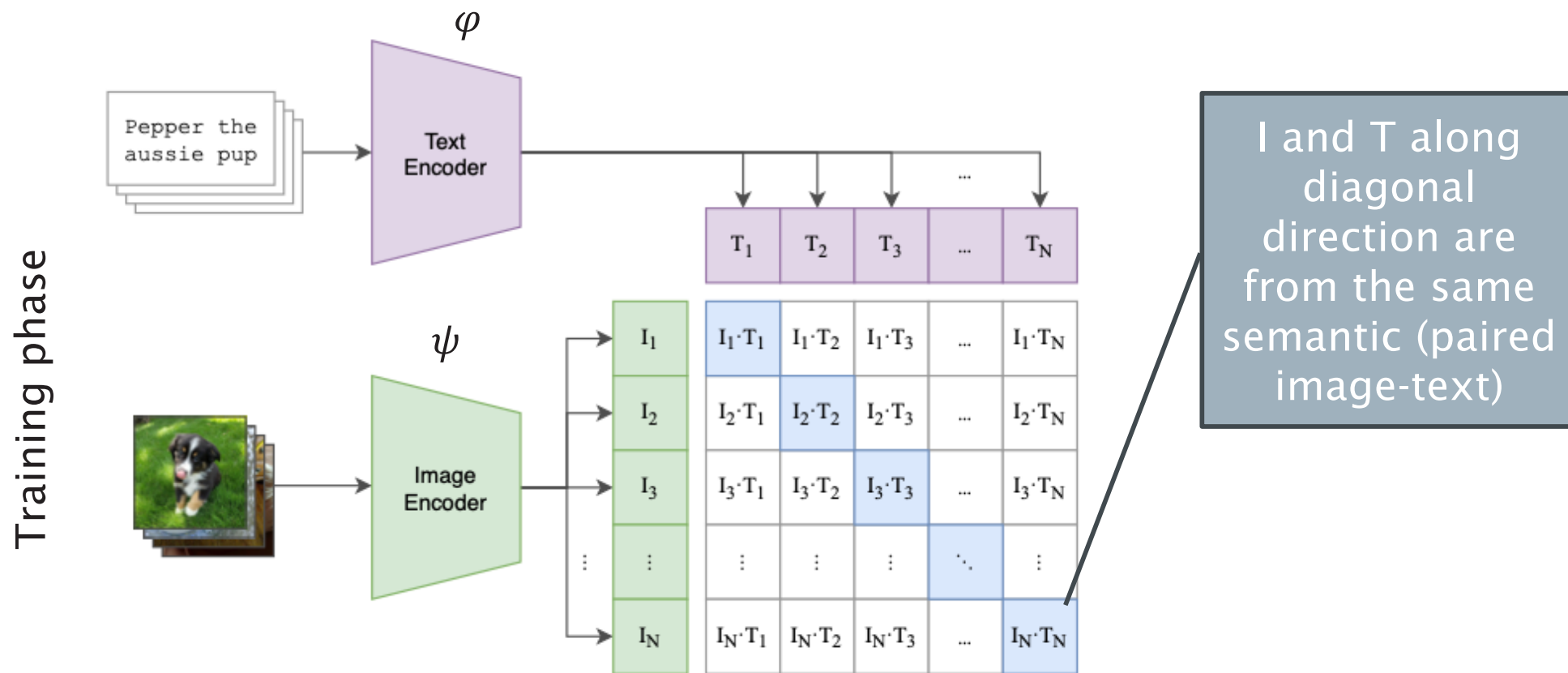
Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)



CLIP uses an abundantly available source of supervision: the text paired with images found across the internet



Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)

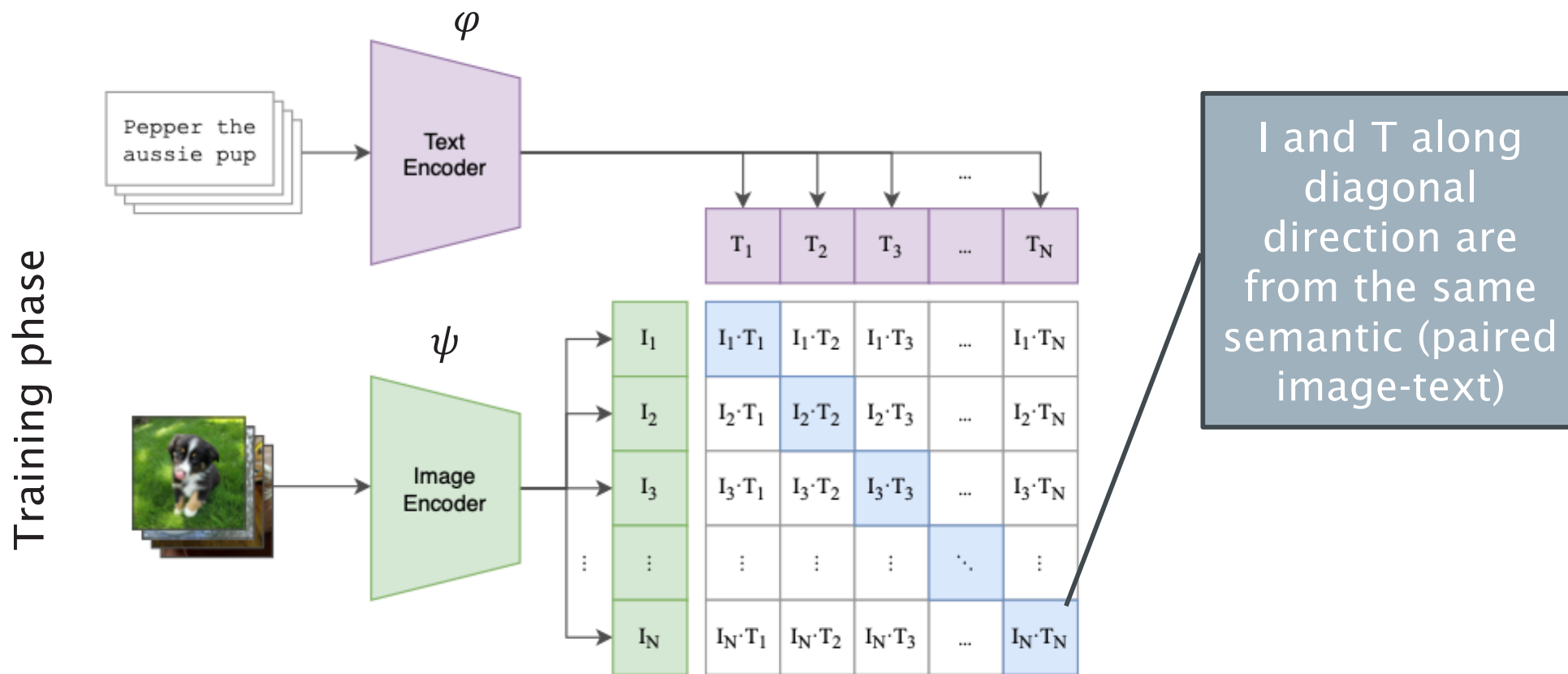


Radford et al., 2021

Text encoder: find the latent features of text with a non-linear mapping $\varphi(X)$, where X indicates the text raw features

Image encoder: find the latent features of text with a non-linear mapping $\psi(Y)$, where Y indicates the image raw features

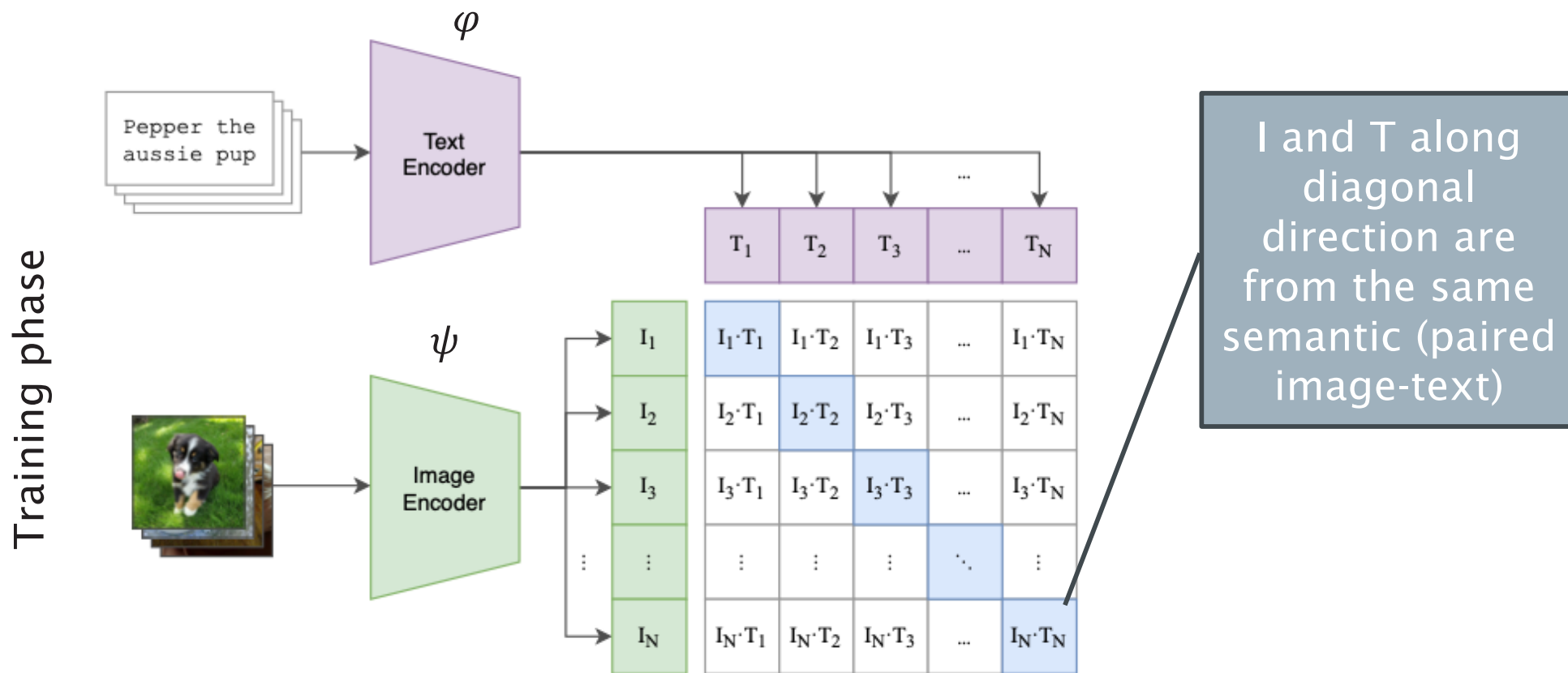
Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)



Radford et al., 2021

Similarity: $A = \varphi(X)\psi(Y)^T$, where X, Y indicate the text and image raw features, φ and ψ are the mapping functions of text/image encoder

Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)



Radford et al., 2021

Supervision for training: $\min \ell(A, G)$, where A, G are the similarity matrix and ground truth, and ℓ is the cross-entropy loss (the lower the closer distance from A to G)

Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)

Numpy-like pseudocode for the core of an implementation of CLIP

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t            - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

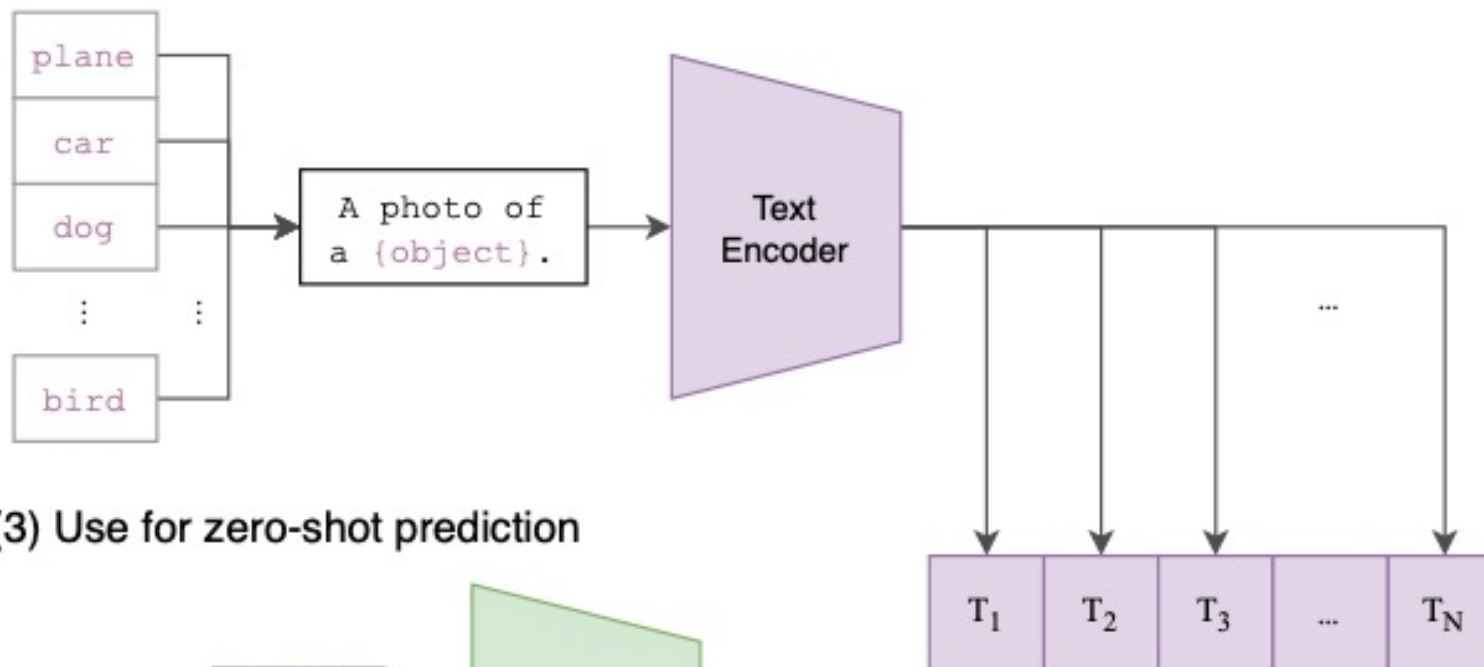
# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

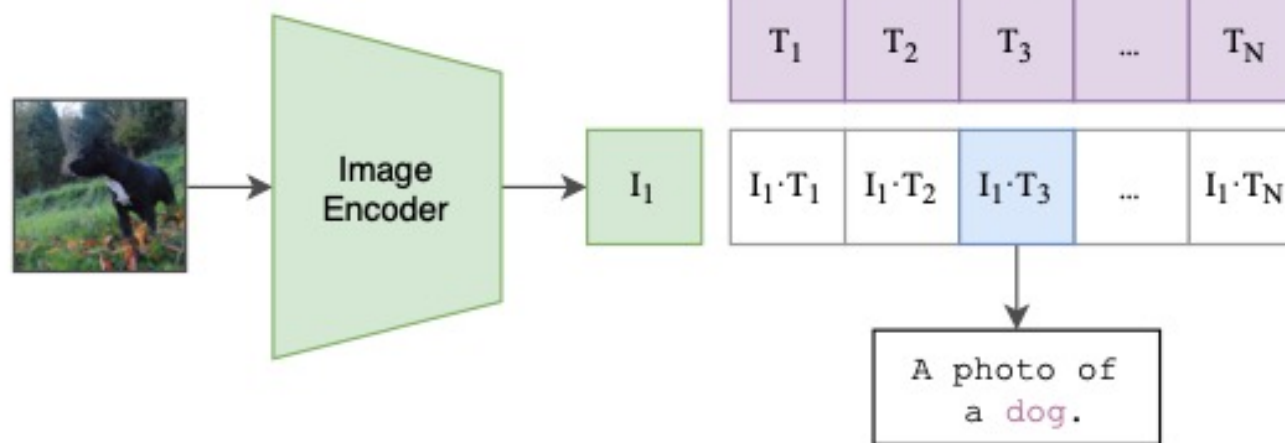
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



Semantic Spaces – Contrastive Language-Image Pre-training (CLIP)

Food101

guacamole (90.1%) Ranked 1 out of 101 labels



✓ a photo of **guacamole**, a type of food.

× a photo of **ceviche**, a type of food.

× a photo of **edamame**, a type of food.

× a photo of **tuna tartare**, a type of food.

× a photo of **hummus**, a type of food.

Youtube-BB

airplane, person (89.0%) Ranked 1 out of 23 labels



✓ a photo of a **airplane**.

× a photo of a **bird**.

× a photo of a **bear**.

× a photo of a **giraffe**.

× a photo of a **car**.

SUN397

television studio (90.2%) Ranked 1 out of 397 labels



✓ a photo of a **television studio**.

× a photo of a **podium indoor**.

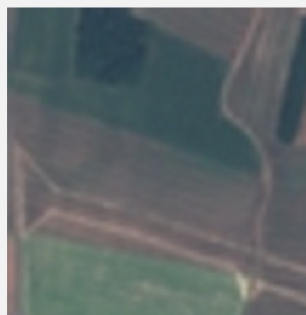
× a photo of a **conference room**.

× a photo of a **lecture room**.

× a photo of a **control room**.

EuroSAT

annual crop land (46.5%) Ranked 4 out of 10 labels



× a centered satellite photo of **permanent crop land**.

× a centered satellite photo of **pasture land**.

× a centered satellite photo of **highway or road**.

✓ a centered satellite photo of **annual crop land**.

× a centered satellite photo of **brushland or shrubland**.

<https://openai.com/research/clip>

Semantic Spaces – Summary

○ **Latent Semantic Analysis (LSA)**

- Shallow Learning Approach: BoW & truncated SVD, Simple and efficient
- Feasible extensions for Multimodal LSA: learns from both image and text data
- Abstract Concepts: represented with linear mixtures of words
- Unconstrained Weights: Weights could be negative, impacting interpretation
- Limited Semantic Interpretation: Topics may lack semantic meaning

○ **Contrastive Language-Image Pre-training (CLIP)**

- Deep Learning Approach: Deep Networks, data- and compute-hungry
- Multimodal Understanding: Learns from both image and text data
- Abstract Concepts: Captures complex semantic relationships
- Challenges in Interpretability: deep learning nature may hinder interpretability